

A generalized three-dimensional Hoek–Brown strength criterion

By

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Summary

Although the Hoek–Brown strength criterion has been widely used in rock mechanics and rock engineering, it does not take account of the influence of the intermediate principal stress. Much evidence, however, has been accumulating to indicate that the intermediate principal stress does influence the rock strength in many instances. Therefore, researchers have developed three-dimensional (3D) versions of the Hoek–Brown strength criterion. In this paper, three existing 3D versions of the Hoek–Brown strength criterion are reviewed and evaluated. The evaluation shows that all of the three 3D versions of the Hoek–Brown strength criterion have limitations. To address the limitations, a generalized 3D Hoek–Brown criterion is proposed by modifying the generalized Hoek–Brown strength criterion. The proposed 3D criterion not only inherits the advantages of the Hoek–Brown strength criterion but can take account of the influence of the intermediate principal stress. At a 2D stress state (triaxial or biaxial), the proposed 3D criterion will simply reduce to the form of the generalized Hoek–Brown strength criterion. To validate the proposed 3D strength criterion, polyaxial or true triaxial compression test data of intact rocks and jointed rock masses has been collected from the published literature. Predictions of the proposed generalized 3D Hoek–Brown strength criterion are in good agreement with the test data for a range of different rock types. The difference of the proposed generalized 3D Hoek–Brown strength criterion from and its advantages over the existing 3D versions of the Hoek–Brown strength criterion are also discussed. It should be noted that the proposed 3D criterion is empirical in nature because it is an extension of the 2D Hoek–Brown strength criterion, which is empirical. Because of the non-convexity of the yield surface for a biaxial stress state, the proposed 3D criterion may have problems with some stress paths.

Keywords: Rock, rock mass, Hoek–Brown strength criterion, 3D strength criterion, intermediate principal stress

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1. Introduction

This paper is an extension of the original paper by Zhang and Zhu (2007) and aims to (1) update the three-dimensional (3D) criterion proposed by Zhang and Zhu (2007) to accommodate the generalized form of the Hoek–Brown criterion, characterized by the exponent “ a ” rather than 0.5; (2) review the earlier versions of the 3D Hoek–Brown strength criterion and compare them with the newly derived generalized 3D Hoek–Brown strength criterion; and (3) validate the new generalized 3D Hoek–Brown strength criterion. The test data of intact rocks and jointed rock masses used in Zhang and Zhu (2007) are re-used for validation and the figures in Zhang and Zhu (2007) are reproduced with modifications to include new information.

The Hoek–Brown criterion has been used widely in rock engineering practice for the following reasons (Priest, 2005):

- The Hoek–Brown criterion has been developed specifically for rock materials and rock masses.
- Input parameters for the Hoek–Brown criterion can be derived from uniaxial testing of the rock material, mineralogical examination, and characterization of rock discontinuities.
- The Hoek–Brown criterion has been applied for almost three decades by practitioners in rock engineering, and has been applied successfully to a wide range of intact and fractured rock types (Hoek and Brown, 1997).

The Hoek–Brown strength criterion is an empirical criterion which was developed by trial and error and is based on the observed behavior of rock masses, model studies to simulate the failure mechanism of jointed rock, and triaxial compression tests of fractured rock (Hoek and Brown, 1980, 1988, 1997; Hoek et al., 2002). The original Hoek–Brown strength criterion is defined by (Hoek and Brown, 1980):

$$\sigma'_1 = \sigma'_3 + \sigma_c \left(m \frac{\sigma'_3}{\sigma_c} + s \right)^{0.5} \quad (1)$$

where σ_c is the unconfined compressive strength of the intact rock; σ'_1 and σ'_3 are, respectively, the major and minor effective principal stresses; and m and s are material constants, where $m = m_i$ and $s = 1$ for intact rock. m_i depends only upon the rock type (texture and mineralogy) and can be selected from Table 1 (Hoek and Brown, 1997; Marinos and Hoek, 2001). This criterion was later updated (Hoek and Brown, 1988) and modified (Hoek et al., 1992) to the current generalized form as follows:

$$\sigma'_1 = \sigma'_3 + \sigma_c \left(m_b \frac{\sigma'_3}{\sigma_c} + s \right)^a \quad (2)$$

where m_b is the value of constant m for the rock mass; and s and a are constants that depend on the characteristics of the rock mass. The parameters m_b , s and a can be estimated from the Geological Strength Index (GSI) as follows (Hoek et al., 2002):

$$m_b = \exp \left(\frac{\text{GSI} - 100}{28 - 14D} \right) m_i \quad (3)$$

$$s = \exp\left(\frac{\text{GSI} - 100}{9 - 3D}\right) \quad (4)$$

$$a = 0.5 + \frac{1}{6}[\exp(-\text{GSI}/15) - \exp(-20/3)] \quad (5)$$

where D is a factor that depends on the degree of disturbance due to blast damage and stress relaxation. Values of D range from 0 for undisturbed in situ rock masses to 1 for very disturbed rock masses. Guidelines for the selection of D can be found in Hoek (2006).

The Hoek–Brown strength criterion, either in the original or the current form, does not take account of the influence of the intermediate principal stress. Much evidence, however, has been accumulating to indicate that the intermediate principal stress does influence the rock strength in many instances (Mogi, 1971; Pan and Hudson, 1988;

Table 1. Values of parameter m_i for different rocks (after Hoek and Brown, 1997; Marinos and Hoek, 2001)

Rock type	Class	Group	Texture			
			Coarse	Medium	Fine	Very fine
Sedimentary	clastic		conglomerate (21 ± 3) ^a	sandstone (17 ± 4)	siltstone (7 ± 2)	claystone (4 ± 2)
			breccia (19 ± 5)		greywacke (18 ± 3)	shale (6 ± 2)
	non-clastic	carbonate	crystalline limestone (12 ± 3)	sparitic limestone (10 ± 2)	micritic limestone (9 ± 2)	marl (7 ± 2)
		evaporite		gypsum (8 ± 2)	anhydrite (12 ± 2)	
		organic				chalk (7 ± 2)
Metamorphic	non-foliated		marble (9 ± 3)	hornfels (19 ± 4)	quartzite 20 ± 3	
				metasandstone (19 ± 3)		
	slightly foliated foliated ^b		migmatite (29 ± 3)	amphibolite (26 ± 6)	gneiss (28 ± 5)	
				schist (12 ± 3)	phyllite (7 ± 3)	slate (7 ± 4)
Igneous	plutonic	light	granite (32 ± 3)	diorite (25 ± 5)		
			granodiorite (29 ± 3)			
		dark	gabbro (27 ± 3)	dolerite (16 ± 5)		
	hypabyssal		norite (20 ± 5)			
			porphyrie (20 ± 5)		diabase (15 ± 5)	peridotite (25 ± 5)
	volcanic	lava		rhyolite (25 ± 5)	dacite (25 ± 3)	obsidian (19 ± 3)
				andesite (25 ± 5)	basalt (25 ± 5)	
		pyroclastic	agglomerate (19 ± 3)	breccia (19 ± 5)	tuff (13 ± 5)	

^a Values in parenthesis are estimates

^b These values are for intact rock specimen tests normal to bedding or foliation. The value of m_i will be significantly different if failure occurs along a weakness plane

Brown, 1993; Lade, 1993; Wang and Kemeny, 1995; Chang and Haimson, 2000; Colmenares and Zoback, 2002; Al-Ajmi and Zimmerman, 2005). Therefore, researchers have developed 3D versions of the Hoek–Brown strength criterion. The 3D versions of the Hoek–Brown strength criterion have advantages over other existing 3D non-Hoek–Brown type criteria in that they use the same input parameters as the widely used Hoek–Brown criterion (Priest, 2005; Zhang and Zhu, 2007).

In this paper, three 3D versions of the Hoek–Brown strength criterion are reviewed and evaluated. The evaluation shows that all of the existing 3D versions of the Hoek–Brown strength criterion have limitations. To address the limitations, a generalized 3D Hoek–Brown criterion is proposed. The proposed 3D criterion is validated using polyaxial or true triaxial compression test data of intact rocks and jointed rock masses collected from the published literature. The difference of the proposed 3D criterion from and its advantages over the existing 3D versions of the Hoek–Brown strength criterion are also discussed.

2. Existing 3D versions of Hoek–Brown strength criterion

In this section, three 3D versions of the Hoek–Brown strength criterion are briefly described and evaluated.

2.1 Pan–Hudson criterion

Pan and Hudson (1988) proposed a 3D version of the original Hoek–Brown strength criterion (Eq. (1)), which is expressed as

$$\frac{3}{\sigma_c} J_2 + \frac{\sqrt{3}}{2} m \sqrt{J_2} - m \frac{I_1}{3} = s \sigma_c \quad (6)$$

where I_1 and J_2 are, respectively, the first stress invariant and the second deviatoric stress invariant defined by

$$I_1 = \sigma'_1 + \sigma'_2 + \sigma'_3 \quad (7)$$

$$J_2 = \frac{(\sigma'_1 - \sigma'_2)^2 + (\sigma'_2 - \sigma'_3)^2 + (\sigma'_3 - \sigma'_1)^2}{6} \quad (8)$$

where σ'_1 , σ'_2 and σ'_3 are, respectively, the major, intermediate and minor effective principal stresses.

2.2 Priest criterion

Priest (2005) developed a 3D version of the generalized Hoek–Brown strength criterion (Eq. (2)) by combining it with the Drucker–Prager criterion which is given by

$$J_2^{1/2} = A J_1 + B \quad (9)$$

where A and B are empirical parameters; and $J_1 = I_1/3$ is the mean effective stress. Since the 3D stress state satisfying the generalized Hoek–Brown strength criterion (Eq. (2)) is $(\sigma'_1, \sigma'_3, \sigma'_3)$, the strategy is to determine parameters A and B for the

Drucker-Prager failure surface that intersects the Hoek–Brown failure point ($\sigma'_1, \sigma'_3, \sigma'_3$). With parameters A and B determined, the strength σ'_{zf} for an element of material subjected to a general state of 3D stresses $\sigma'_x \neq \sigma'_y \neq \sigma'_{zf}$ can then be predicted. This process is equivalent to the well known process identifying the Drucker-Prager parameters that give the circumscribed fit for the Coulomb strength criterion. The approach for determining parameters A and B is outlined below.

Consider conditions of unconfined compression, failure occurs at unconfined compressive strength σ_{cm} which can be determined from Eq. (2) as

$$\sigma_{cm} = s^a \sigma_c \quad (10)$$

(In the original paper, Priest (2005) only considered the unconfined compressive strength σ_c for the intact rock. For the generalized Hoek–Brown strength criterion, however, one should consider the unconfined compressive strength σ_{cm} for the rock mass). Under unconfined compression conditions

$$J_1 = \frac{\sigma_{cm}}{3} \quad (11)$$

$$J_2 = \frac{\sigma_{cm}^2}{3} \quad (12)$$

Substituting Eqs. (11) and (12) into Eq. (9), one obtains

$$B = \frac{\sigma_{cm}(\sqrt{3} - A)}{3} \quad (13)$$

Combining Eqs. (9) and (13), one has

$$A = \frac{3\sqrt{J_2} - \sigma_{cm}\sqrt{3}}{3J_1 - \sigma_{cm}} \quad (14)$$

The mean effective stresses J_1 corresponding to the Hoek–Brown and Drucker-Prager criteria are respectively given by

$$J_{1HB} = \frac{\sigma'_1 + 2\sigma'_3}{3} \quad (15)$$

$$J_{1DP} = \frac{\sigma'_x + \sigma'_y + \sigma'_{zf}}{3} \quad (16)$$

For the two strength criteria to be compatible, one requires $J_{1HB} = J_{1DP} = J_1$, i.e.

$$J_1 = \frac{\sigma'_1 + 2\sigma'_3}{3} \quad (17)$$

$$J_1 = \frac{\sigma'_x + \sigma'_y + \sigma'_{zf}}{3} \quad (18)$$

The second deviatoric stress invariant J_2 for the Drucker-Prager failure under effective principal stress state $\sigma'_x \neq \sigma'_y \neq \sigma'_{zf}$ is given by

$$J_2 = \frac{(\sigma'_x - \sigma'_y)^2 + (\sigma'_y - \sigma'_{zf})^2 + (\sigma'_{zf} - \sigma'_x)^2}{6} \quad (19)$$

The radial distances R from the central mean stress point in the π -plane to the failure surfaces corresponding to the Hoek–Brown and Drucker–Prager criteria are respectively given by

$$R_{HB} = \sqrt{(\sigma'_1 - J_1)^2 + 2(\sigma'_3 - J_1)^2} \quad (20)$$

$$R_{DP} = \sqrt{(\sigma'_x - J_1)^2 + (\sigma'_y - J_1)^2 + (\sigma'_{zf} - J_1)^2} \quad (21)$$

In order for the two failure surfaces to intersect, one requires $R_{HB} = R_{DP}$, i.e.

$$\sqrt{(\sigma'_1 - J_1)^2 + 2(\sigma'_3 - J_1)^2} = \sqrt{(\sigma'_x - J_1)^2 + (\sigma'_y - J_1)^2 + (\sigma'_{zf} - J_1)^2} \quad (22)$$

In summary, there are seven independent Eqs. (2), (13), (14), (17), (18), (19) and (22) which contain the following seven unknowns:

- σ'_1 and σ'_3 – Stress values in the 2D Hoek–Brown criterion
- A and B – Drucker–Prager parameters
- J_1 and J_2 – Mean effective stress and second deviatoric stress invariant
- σ'_{zf} – Yield stress for the specified principal stresses σ'_x and σ'_y

So it is possible to solve the seven equations for the seven unknowns listed above. Priest (2005) proposed a numerical method for solving these equations on a spreadsheet. The following is an alternate way for solving these equations:

1. Specify the input data σ_c , σ_{cm} , m_b , s , a , σ'_x and σ'_y .
2. Assume initial trial value for σ'_3 . For convenience, these unknown stress values can initially be set to the value of σ_{cm} .
3. Calculate the following values
 - σ'_1 from Eq. (2)
 - σ'_{zf} from Eqs. (17) and (18)
 - J_1 from Eqs. (17) or (18)
 - J_2 from Eq. (19)
 - R_{HB} and R_{DP} respectively from Eqs. (20) and (21)
4. Check the error defined as follows

$$\varepsilon_c = |R_{HB} - R_{DP}| \quad (23)$$
5. Change the value of σ'_3 and repeat steps 3 and 4 so that the error term ε_c be minimized.
6. Calculate A and B respectively from Eqs. (14) and (13).

2.3 Zhang–Zhu Criterion

Zhang and Zhu (2007) proposed a 3D version of the original Hoek–Brown strength criterion (Eq. (1)), which is expressed as

$$\frac{9}{2\sigma_c} \tau_{oct}^2 + \frac{3}{2\sqrt{2}} m \tau_{oct} - m \sigma'_{m,2} = s \sigma_c \quad (24)$$

where τ_{oct} and $\sigma'_{m,2}$ are, respectively, the octahedral shear stress and the effective mean stress defined by

$$\tau_{\text{oct}} = \frac{1}{3} \sqrt{(\sigma'_1 - \sigma'_2)^2 + (\sigma'_2 - \sigma'_3)^2 + (\sigma'_3 - \sigma'_1)^2} \quad (25)$$

$$\sigma'_{m,2} = \frac{\sigma'_1 + \sigma'_3}{2} \quad (26)$$

1.4 Evaluation of existing 3D versions of Hoek–Brown strength criterion

Both Pan–Hudson and Zhang–Zhu criteria are the 3D version of the original Hoek–Brown strength criterion (Eq. (1)). Since $J_2 = (3/2)\tau_{\text{oct}}^2$, the Pan–Hudson criterion (Eq. (6)) can be rewritten as

$$\frac{9}{2\sigma_c} \tau_{\text{oct}}^2 + \frac{3}{2\sqrt{2}} m \tau_{\text{oct}} - m J_1 = s \sigma_c \quad (27)$$

One can see that the only difference between the Zhang–Zhu criterion (Eq. (24)) and the Pan–Hudson criterion (Eq. (6)) is the mean effective stress term.

For $\sigma'_1 = \sigma'_2$ or $\sigma'_2 = \sigma'_3$, $\tau_{\text{oct}} = \sqrt{2}(\sigma'_1 - \sigma'_3)/3$ and Eq. (24) will simply reduce to Eq. (1). So the Zhang–Zhu criterion will predict the same strength as the original Hoek–Brown strength criterion (Eq. (1)) at both biaxial and triaxial stress states. When $\sigma'_1 = \sigma'_2$ or $\sigma'_2 = \sigma'_3$, however, the Pan–Hudson criterion (Eq. (6)) will not reduce to the form of the original Hoek–Brown criterion (Eq. (1)). So the Pan–Hudson criterion will not predict the same strength as the original Hoek–Brown criterion at either biaxial or triaxial stress state.

The Priest criterion is a 3D version of the generalized Hoek–Brown strength criterion (Eq. (2)). Since it is derived by identifying the Drucker–Prager parameters that give the circumscribed fit for the Hoek–Brown criterion, the Priest criterion will predict the same strength as the generalized Hoek–Brown strength criterion at triaxial stress state ($\sigma'_2 = \sigma'_3$) but not at biaxial stress state ($\sigma'_1 = \sigma'_2$). Another limitation of the Priest criterion is that a numerical procedure is required to derive the parameters of the criterion.

Figure 1 shows the π -plane plot of the Hoek–Brown criterion and the above three 3D versions of the Hoek–Brown criterion at $J_1 = 100$ MPa for a rock mass with $\sigma_c = 50$ MPa, $m = 5$, $s = 0.01$, and $a = 0.5$. Only the Zhang–Zhu criterion predicts the same strength as the Hoek–Brown criterion at both triaxial stress state $\sigma'_2 = \sigma'_3$ (points T in the figure) and biaxial stress state $\sigma'_1 = \sigma'_2$ (points B in the figure). The Pan–Hudson criterion under-predicts the strength at triaxial stress state and over-predicts the strength at biaxial stress state. The Priest criterion predicts the same strength as the Hoek–Brown criterion at triaxial stress state but over-predicts the strength at biaxial stress state. It is noted that the Zhang–Zhu criterion envelope is concave at points B. Because of the non-convexity, the proposed criterion may have problems with some stress paths. A non-convex (or concave) criterion, so far, has only been proposed by Yu (2002) and Yu et al. (2002).

Table 2 shows the ratios $\tau_{\text{oct}}(\text{Pan–Hudson})/\tau_{\text{oct}}(\text{Hoek–Brown})$ and $\tau_{\text{oct}}(\text{Priest})/\tau_{\text{oct}}(\text{Hoek–Brown})$ at $J_1 = 100$ MPa for three different rock masses. The difference between $\tau_{\text{oct}}(\text{Pan–Hudson})$ or $\tau_{\text{oct}}(\text{Priest})$ and $\tau_{\text{oct}}(\text{Hoek–Brown})$ increases as the rock

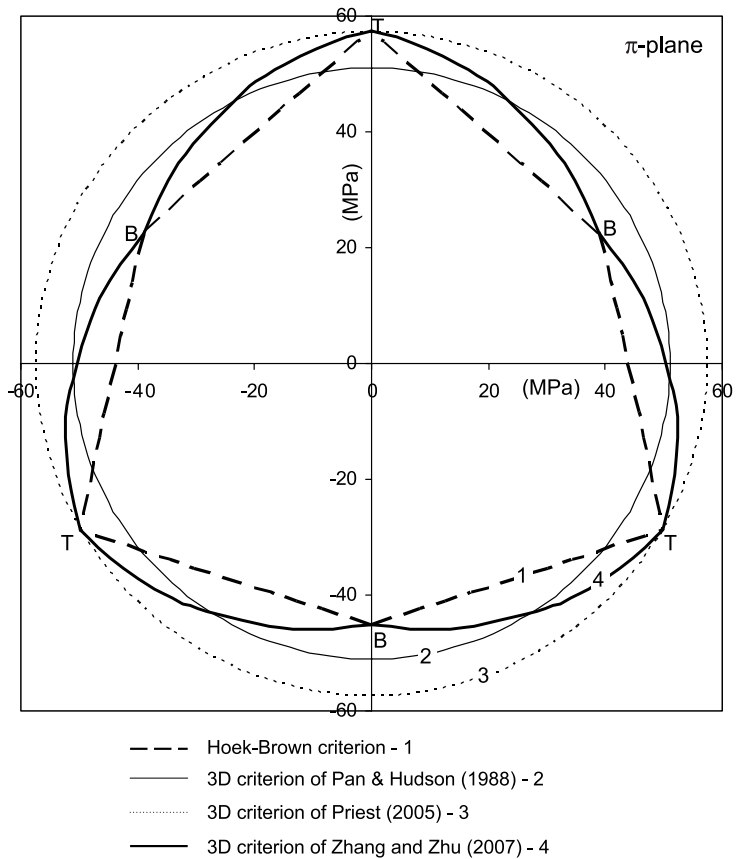


Fig. 1. π -plane plot of Hoek–Brown criterion and 3D criteria of Pan and Hudson (1988), Priest (2005) and Zhang and Zhu (2007) at $J_1 = 100$ MPa for a rock mass with $\sigma_c = 50$ MPa, $m = 5$, $s = 0.01$ and $a = 0.5$

Table 2. Ratios of $\tau_{oct}(\text{Pan–Hudson})$ to $\tau_{oct}(\text{Hoek–Brown})$ and $\tau_{oct}(\text{Priest})$ to $\tau_{oct}(\text{Hoek–Brown})$ at $J_1 = 100$ MPa for three different rock masses

Rock Mass ¹	σ_c (MPa)	m_i	GSI	$\frac{\tau_{oct}(\text{Pan–Hudson})}{\tau_{oct}(\text{Hoek–Brown})}$		$\frac{\tau_{oct}(\text{Priest})}{\tau_{oct}(\text{Hoek–Brown})}$
				Triaxial state	Biaxial state	Biaxial state
Very good quality	150	25	75	0.78	1.24	1.60
Average	80	12	50	0.90	1.10	1.22
Very poor quality	20	8	30	0.97	1.03	1.06

¹ Rock mass quality based on Hoek (2006).

mass becomes stronger. Since the Hoek–Brown criterion has been applied successfully to a wide range of intact and fractured rock types, it is essential for a 3D criterion to predict the same strength as the Hoek–Brown criterion at both triaxial and biaxial

stress states because an accurate 3D criterion should also accurately predict the strength at 2D stress state. So the Zhang–Zhu criterion can be applied to all rock masses to which the original Hoek–Brown criterion was applied; however, the Pan–Hudson and Priest criteria only apply approximately to relatively weaker rock masses (i.e., rock mass with small σ_c and/or small m and s).

3. A generalized three-dimensional Hoek–Brown strength criterion

The Zhang–Zhu criterion is a 3D version of the original Hoek–Brown criterion (Eq. (1)). So it might not be applicable to some of the rock masses to which the generalized Hoek–Brown criterion (Eq. (2)) applies. Therefore, the Zhang–Zhu criterion is modified so that it is applicable to all of the rock masses to which the generalized Hoek–Brown criterion applies.

Considering the differences between the original Hoek–Brown criterion (Eq. (1)) and the generalized Hoek–Brown criterion (Eq. (2)), the Zhang–Zhu criterion (Eq. (24)) is modified as follows

$$\frac{1}{\sigma_c^{(1/a-1)}} \left(\frac{3}{\sqrt{2}} \tau_{\text{oct}} \right)^{1/a} + \frac{m_b}{2} \left(\frac{3}{\sqrt{2}} \tau_{\text{oct}} \right) - m_b \sigma'_{m,2} = s \sigma_c \quad (28)$$

When $a = 0.5$, Eq. (28) will be simplified to Eq. (24). Derivation of the generalized 3D criterion (Eq. 28) is based on the following general strength criterion suggested by Mogi (1971):

$$\tau_{\text{oct}} = f(\sigma'_{m,2}) \quad (29)$$

Equation (29) is based on a number of polyaxial compressive tests indicating that the failure plane strikes in the direction of the intermediate principal stress. Al-Ajmi and Zimmerman (2005) validated assumption (29) by applying it to polyaxial test data for different rocks.

For $\sigma'_1 = \sigma'_2$ or $\sigma'_2 = \sigma'_3$, Eq. (28) will reduce to Eq. (2). So the generalized Hoek–Brown strength criterion is just a 2D version of the proposed generalized 3D strength criterion, or the proposed criterion is a *true* 3D version of the generalized Hoek–Brown criterion.

4. Validation of the proposed criterion

In this section, the proposed generalized 3D Hoek–Brown strength criterion is validated by applying it to both intact rocks and jointed rock masses.

4.1 Application to intact rocks

For intact rocks, $m_b = m_i$, $s = 1$, $a = 0.5$ and the proposed generalized 3D Hoek–Brown strength criterion is the same as the Zhang–Zhu criterion which has been validated using test data of five rock types (Zhang and Zhu, 2007). The first rock type is Amphibolite from the KTB site and was tested by Chang and Haimson (2000). The

Table 3. Polyaxial test data for KTB Amphibolite (Chang and Haimson, 2000; Colmenares and Zoback, 2002; Al-Ajmi and Zimmerman, 2005)

σ'_1 (MPa)	σ'_2 (MPa)	σ'_3 (MPa)
165	0	0
346	79	0
291	149	0
347	197	0
267	229	0
410	30	30
479	60	30
599	100	30
652	200	30
571	249	30
637	298	30
702	60	60
750	88	60
766	103	60
745	155	60
816	199	60
888	249	60
828	299	60
887	347	60
954	399	60
815	449	60
868	100	100
959	164	100
1001	199	100
945	248	100
892	269	100
1048	300	100
1058	349	100
1155	442	100
1118	597	100
1147	150	150
1065	198	150
1112	199	150
1176	249	150
1431	298	150
1326	348	150
1169	399	150
1284	448	150
1265	498	150
1262	642	150

second rock type is Westerly granite and was tested by Haimson and Chang (2000). The other three rock types are Dunham dolomite, Apache Leap tuff and Mizuho trachyte, and were tested respectively by Mogi (1971), Wang and Kemeny (1995) and Mogi (1971). The test types for these 5 rocks are uniaxial, triaxial and true triaxial and the test data is listed in Tables 3–7. The unconfined compressive strength values of these rocks are listed in Table 8. Although these test data have also been used by other researchers (e.g., Colmenares and Zoback, 2002; Al-Ajmi and Zimmerman, 2005) to study the strength criteria of rocks, it is important to note that different test conditions may have unknown effects. This limits the generalization of the validation to some extent.

Table 4. Polyaxial test data for Westerly granite (Haimson and Chang, 2000; Al-Ajmi and Zimmerman, 2005)

σ'_1 (MPa)	σ'_2 (MPa)	σ'_3 (MPa)
201	0	0
306	40	0
301	60	0
317	80	0
304	100	0
231	2	2
300	18	2
328	40	2
359	60	2
353	80	2
355	100	2
430	20	20
529	40	20
602	60	20
554	62	20
553	61	20
532	79	20
575	100	20
567	114	20
601	150	20
638	202	20
605	38	38
620	38	38
700	57	38
733	78	38
720	103	38
723	119	38
731	157	38
781	198	38
747	60	60
811	90	60
821	114	60
860	180	60
861	249	60
889	77	77
954	102	77
992	142	77
998	214	77
1005	310	77
1012	100	100
1103	165	100
1147	167	100
1155	216	100
1195	259	100
1129	312	100

Figures 2–6 show the results from Zhang and Zhu (2007) with the predictions from the Priest criterion added. In these figures, the lower bound and upper bound curves for each criterion are obtained by, respectively, using the lower bound and upper bound values of m_i for each specific rock (see Table 1). The curve between the lower bound and upper bound curves is the one that best fits the test data. The best fitting curve is obtained using the method of least squares (LS), i.e., by

Table 5. Polyaxial test data for Dunham dolomite (Mogi, 1971; Al-Ajmi and Zimmerman, 2005)

σ'_1 (MPa)	σ'_2 (MPa)	σ'_3 (MPa)
257	0	0
400	25	25
474	68	25
500	91	25
553	135	25
574	177	25
594	232	25
544	269	25
488	45	45
562	100	45
586	124	45
607	159	45
639	183	45
671	241	45
670	263	45
622	293	45
568	65	65
636	113	65
642	152	65
687	208	65
684	259	65
725	306	65
700	390	65
624	85	85
682	126	85
718	150	85
743	230	85
771	300	85
818	371	85
798	440	85
679	105	105
776	165	105
784	202	105
804	265	105
822	331	105
839	351	105
820	411	105
863	266	105
724	125	125
823	186	125
859	241	125
862	288	125
893	359	125
942	411	125
918	458	125
887	510	125
892	254	145
929	292	145
924	319	145
922	342	145
1016	387	145
1003	404	145
953	451	145

Table 6. Polyaxial test data for Apache Leap tuff (Wang and Kemeny, 1995)

σ'_1 (MPa)	σ'_2 (MPa)	σ'_3 (MPa)
141.9	0	0
141.9	0	0
155.2	0	0
152.9	0	0
169.5	0	0
120.6	0	0
145.5	0	0
131.3	0	0
152.6	0	0
161.5	0	0
181.8	3.4	3.4
197.0	6.9	6.9
240.4	10.3	10.3
248.9	13.8	13.8
250.0	13.8	13.8
265.3	17.2	17.2
242.3	20.7	20.7
308.4	24.1	24.1
329.8	27.6	27.6
356.7	31.0	31
360.2	34.5	34.5
396.5	37.9	37.9
419.5	41.4	41.4
456.1	44.8	44.8
434.1	48.3	48.3
464.9	55.2	55.2
190.1	18.4	0
227.8	36.8	0
241.1	55.2	0
286	73.5	0
325.9	110.3	0
341.3	92.3	0
333.7	79.9	0
272.6	61.5	0
276.6	50.7	0
218.1	30.7	0
180.8	15.4	0
164.7	14.8	0
221.4	29.6	0
247.1	44.5	0
286.6	59.3	0
349.6	88.9	0
170.1	8.6	3.4
223.9	17.2	6.9
200.2	25.7	10.3
255.6	34.5	13.8
241.9	43.1	17.2
289.3	51.7	20.7
373.9	60.3	24.1
419.1	68.9	27.6
170.9	8.6	6.9
242.3	12.9	10.3
201.7	17.2	13.8
260.1	21.5	17.2
244.8	25.9	20.7
328.8	30.2	24.1
295.9	34.5	27.6
340.3	38.8	31.0

Table 7. Polyaxial test data for Mizuho trachyte (Mogi, 1971; Al-Ajmi and Zimmerman, 2005)

σ'_1 (MPa)	σ'_2 (MPa)	σ'_3 (MPa)
100	0	0
193	15	15
253	30	30
300	45	45
314	55	45
326	71	45
333	96	45
349	142	45
361	214	45
365	289	45
351	332	45
339	60	60
352	91	60
383	142	60
396	191	60
404	229	60
400	271	60
383	331	60
365	75	75
400	114	75
417	153	75
438	229	75
439	300	75
424	343	75
451	391	75
419	100	100
460	137	100
489	186	100
494	274	100
522	382	100
513	411	100

Table 8. Values of σ_c , m_i and s used for predicting three-dimensional strength of 5 intact rocks

Rock	σ_c (MPa)	m_i		s
		Range	Best fitting	
KTB Amphibolite	165.0	26 ± 6^1	31	1
Westerly granite	201.0	32 ± 3^1	35	1
Dunham dolomite	257.0	9 ± 3^1	10	1
Apache Leap tuff	147.3	13 ± 5^1	18	1
Mizuho trachyte	100.0	13 ± 4^2	11	1

¹ From Table 1.

² 13 is the estimated value by best-fitting the original Hoek–Brown criterion to the triaxial test data and 4 is simply selected to be the medium of the deviation values 2, 3, 4, 5 and 6 in Table 1.

finding the value of m_i so that the error defined below is the minimum (Zhang and Zhu, 2007):

$$\text{Error} = \sum_{i=1}^n (\tau_{\text{oct,experimental}} - \tau_{\text{oct,predicted}})_i^2 \tag{30}$$

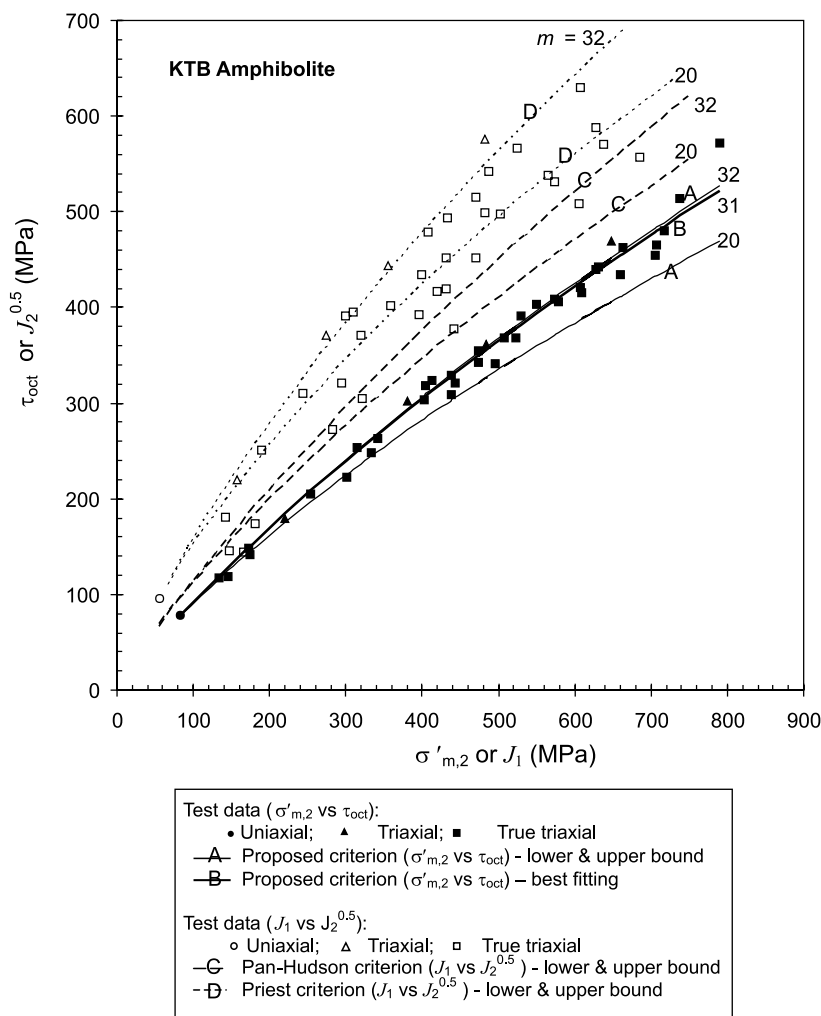


Fig. 2. Comparison of proposed criterion, Pan–Hudson criterion and Priest criterion with test data – KTB amphibolite

where n is the number of test data points. For example, for KTB Amphibolite, $m_i = 26 \pm 6$ according to Table 1. So the lower bound and upper bound curves in Fig. 2 correspond respectively to $m_i = 20$ and 32. The best fitting curve for the proposed criterion corresponds to $m_i = 31$ which is within the range of $m_i = 20$ –32 for amphibolite.

It can be seen from Figs. 2–6 that, for the proposed generalized 3D Hoek–Brown strength criterion, the test data points either fall inside the band formed by the bounding curves or are very close to the bounding curves and the values of m_i corresponding to the best fit are within the range or equal to the upper bound value of m_i for the specific rock (see also Table 8).

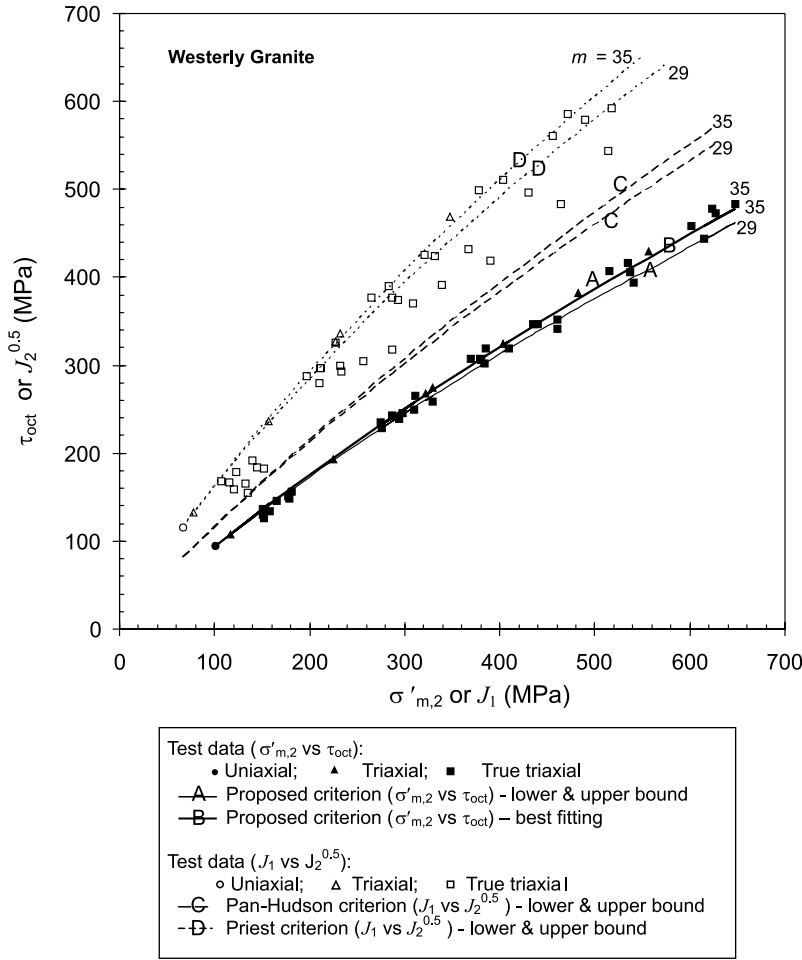


Fig. 3. Comparison of proposed criterion, Pan-Hudson criterion and Priest criterion with test data – Westerly granite

The Pan-Hudson criterion under-predicts the strength for all rocks with the exception only for Mizuho trachyte (which has the lowest σ_c of the five rocks) at high mean effective stress level. The Priest criterion, however, over-predicts the strength for all rocks with the exception only for Apache Leap tuff. Since most of the test data points for the Pan-Hudson criterion and many of the test points for the Priest criterion are out of the band formed by the bound curves, no best fitting analyses are conducted for these two criteria.

4.2 Application to jointed rock masses

The proposed generalized 3D Hoek-Brown strength criterion can also be applied to isotropic jointed rock masses as the generalized Hoek-Brown strength criterion.

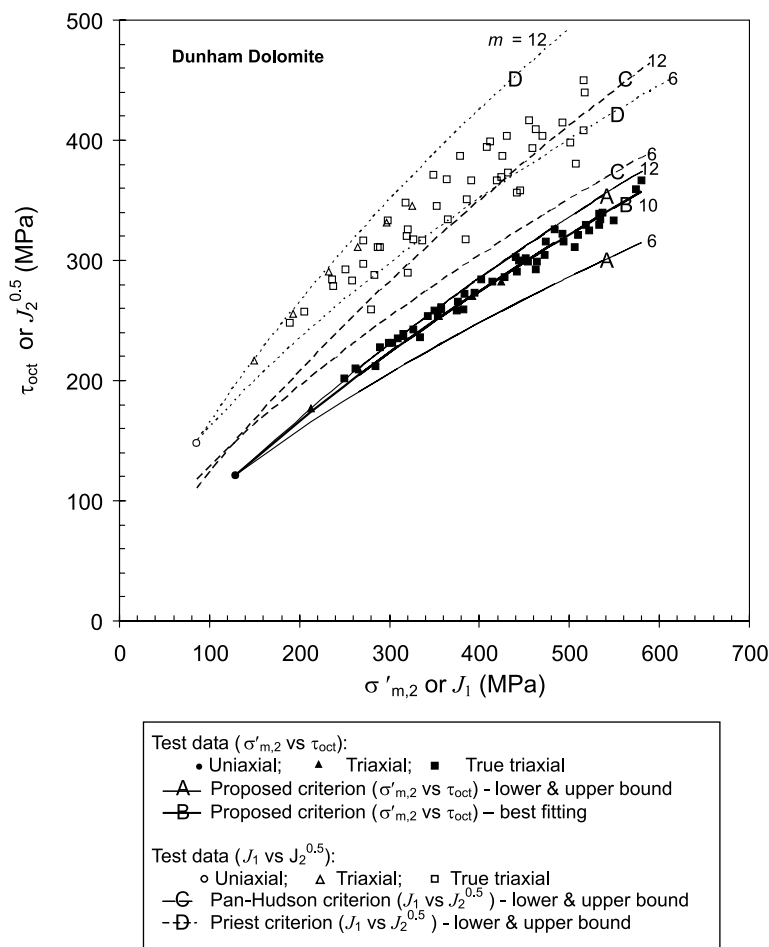


Fig. 4. Comparison of proposed criterion, Pan–Hudson criterion and Priest criterion with test data – Dunham dolomite

Although polyaxial or true triaxial compression tests of jointed rock masses have been conducted by a number of investigators, most of these tests have been conducted on anisotropic jointed specimens to study the influence of the intermediate principal stress on the strength of anisotropic jointed rock masses (Zhang and Zhu, 2007). In the following, the proposed generalized 3D Hoek–Brown strength criterion is applied to analyze the polyaxial test data of jointed biotite granite at the Kurobe IV arch dam site in Japan (Müller-Salzburg and Ge, 1983; Brown, 1993).

Figure 7(a) shows the test block Q_2 which was 2.8 m long in the σ'_1 direction, 2.8 m high in the σ'_2 direction and 1.4 m wide in the σ'_3 direction, giving it a total volume about 11 m^3 . The jointed biotite granite was described as “somewhat decomposed” and the unconfined compressive strength of cores of granite taken

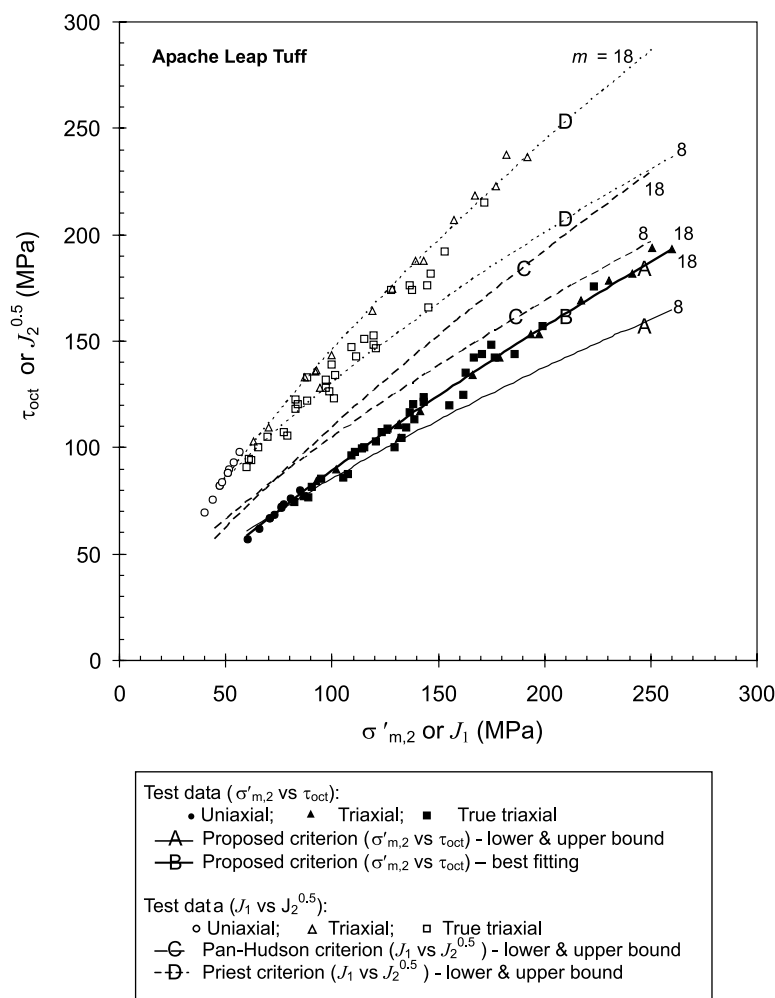


Fig. 5. Comparison of proposed criterion, Pan–Hudson criterion and Priest criterion with test data – Apache Leap tuff

from the test location was 23.0 MPa. The test was conducted by increasing and cycling σ'_1 with σ'_2 and σ'_3 held constant at 0.70 and 0.12 MPa, respectively. Figure 7(b) shows the plot of deformation measured in the σ'_1 direction as σ'_1 was increased and cycled (for the numbers of alternate load cycles, AL, indicated in the figure). From the stress-deformation curves, a peak or failure stress (F) was identified (Müller-Salzburg and Ge, 1983; Brown, 1993). The failure stress was 13.0 MPa which is much lower than the unconfined compressive strength of the corresponding intact rock.

According to Table 1, $m_i = 29$ –35 for granite. For simplicity, an average value of $m_i = 32$ is selected for the analysis. Since the value of GSI is unknown for the tested rock mass, a series of values of GSI is assumed for analyses to create a plot

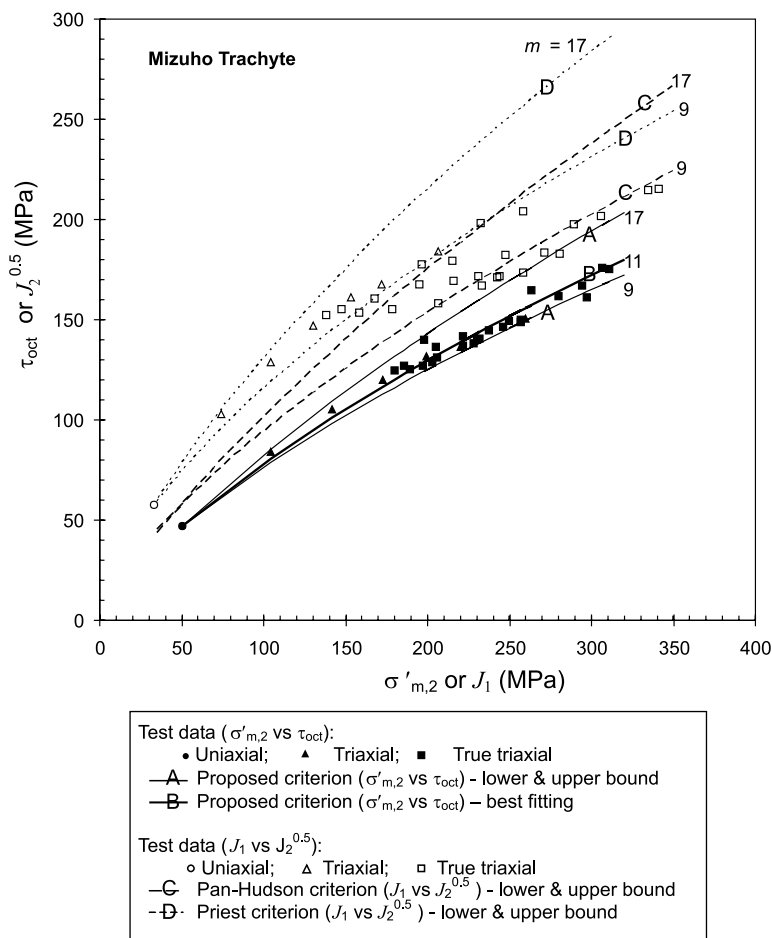


Fig. 6. Comparison of proposed criterion, Pan–Hudson criterion and Priest criterion with test data – Mizuho trachyte

of the failure major principal stress σ'_1 versus GSI at constant σ'_2 and σ'_3 respectively of 0.70 and 0.12 MPa. Assuming that the tested rock mass is undisturbed, i.e., $D=0$, m_b , s and a can be calculated from Eqs. (3) to (5) for an assumed GSI and the corresponding 3D failure major principal stress σ'_1 can be determined from Eq. (28). Figure 8 shows the variation of the obtained failure major principal stress σ'_1 with GSI. For comparison, the failure major principal stress σ'_1 calculated based on the Hoek–Brown strength criterion and the Pan–Hudson, Priest and Zhang–Zhu criteria are also shown in the figure. The Pan–Hudson criterion significantly under-predicts the failure major principal stress σ'_1 . The Zhang–Zhu criterion predicts essentially the same failure major principal stress σ'_1 as the proposed criterion when $GSI > 50$. This is because when $GSI > 50$, a is very close to 0.5 which is what used in the Zhang–Zhu criterion. The failure

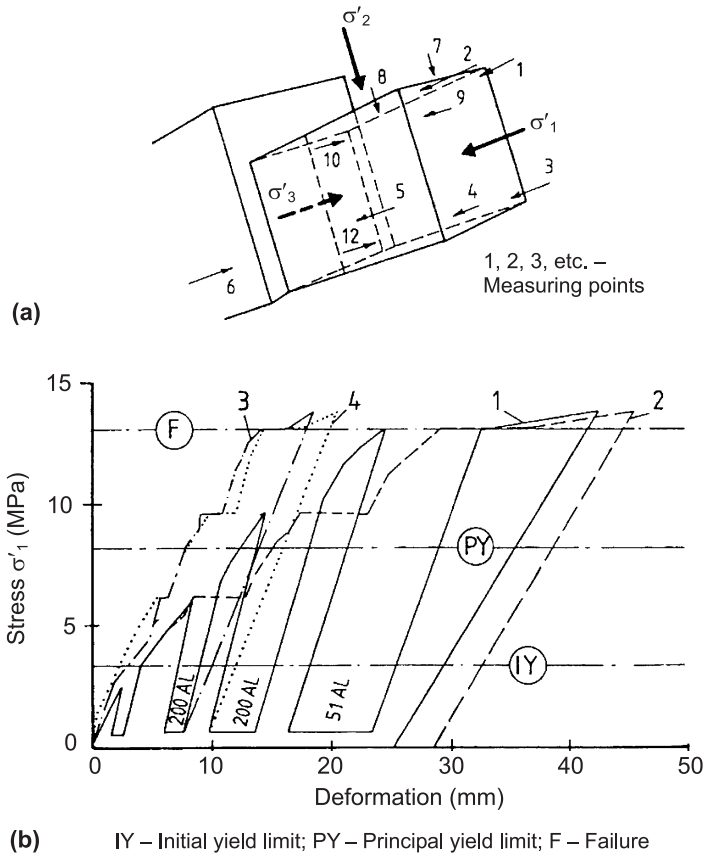


Fig. 7. Polyaxial compression test, Kurobe IV arch dam site, Japan: (a) Test block Q_2 and (b) Major principal compressive stress-deformation curves (after Müller-Salzburg and Ge, 1983)

major principal stress σ'_1 from both the Priest criterion and the proposed criterion is higher than the failure major principal stress σ'_1 from the Hoek–Brown strength criterion, clearly showing the influence of the intermediate principal stress. The measured failure stress σ'_1 of 13.0 MPa corresponds to GSI about 75 (good quality rock mass), GSI about 81 (good quality rock mass) and GSI about 86 (very good quality rock mass) respectively for the Priest criterion, the proposed criterion and the Hoek–Brown criterion. Since the jointed biotite granite was described as “somewhat decomposed” for test block Q_2 , the tested rock mass should be more likely in good quality than in very good quality. As discussed in the previous sections, the Priest criterion over-predicts the strength at true triaxial conditions. So the GSI of the tested rock mass is more likely about 81 as estimated from the proposed criterion than about 75 from the Priest criterion. Considering the uncertainty of the GSI value for this test, future true triaxial tests on rock masses with detailed characterization are recommended for validating the proposed criterion.

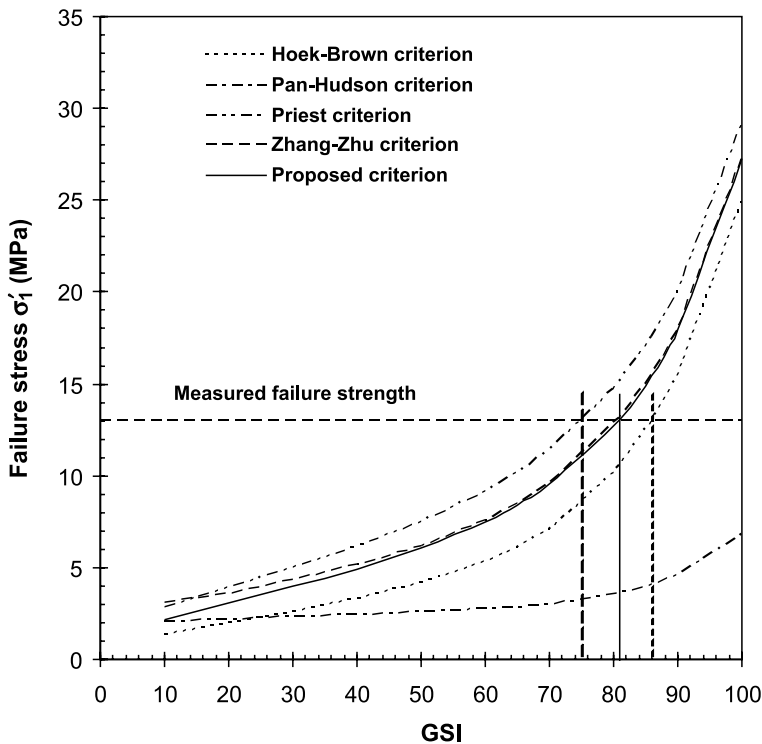


Fig. 8. Variation of failure major principal stress with GSI respectively for the proposed criterion, the Hoek–Brown criterion, and the Pan–Hudson, Priest and Zhang–Zhu criteria

5. Summary and conclusions

This paper has proposed a generalized 3D Hoek–Brown strength criterion that overcomes the limitations of both the Hoek–Brown criterion and the existing 3D versions of the Hoek–Brown criterion. The contents and conclusions can be summarized as follows:

1. The Hoek–Brown strength criterion has been widely used in rock mechanics and rock engineering. The Hoek–Brown strength criterion, however, does not take account of the influence of the intermediate principal stress, although much evidence has been accumulating to indicate that the intermediate principal stress does influence the rock strength.
2. Three 3D versions of the Hoek–Brown strength criterion have been proposed by different researchers to consider the influence of the intermediate principal stress; but all of them have limitations:
 - a. The 3D version of the original Hoek–Brown strength criterion proposed by Pan and Hudson (1988) does not reduce to the form of the original Hoek–Brown criterion at either triaxial or biaxial stress state. Compared with the Hoek–Brown criterion, the Pan–Hudson criterion under-predicts the strength at triaxial stress state but over-predicts the strength at biaxial stress state. The other limi-

- tation of the Pan–Hudson criterion is that it may not be applicable to some of the rock masses to which the generalized Hoek–Brown criterion applies.
- b. The 3D version of the generalized Hoek–Brown strength criterion proposed by Priest (2005) does not reduce to the form of the generalized Hoek–Brown criterion at a biaxial stress state. Compared with the Hoek–Brown criterion, the Priest criterion over-predicts the strength at a biaxial stress state. The other limitation of the Priest criterion is that a numerical procedure is required to determine the parameters of the criterion.
 - c. The 3D version of the original Hoek–Brown strength criterion proposed by Zhang and Zhu (2007) reduces to the form of the original Hoek–Brown criterion at both triaxial and biaxial stress states; but, as the Pan–Hudson criterion, it may not be applicable to some of the rock masses to which the generalized Hoek–Brown criterion applies.
3. To overcome the limitations of both the Hoek–Brown criterion and the existing 3D versions of the Hoek–Brown criterion, a generalized 3D Hoek–Brown strength criterion is proposed. The proposed 3D strength criterion not only inherits the advantages of the Hoek–Brown strength criterion but can take account of the influence of the intermediate principal stress. The proposed criterion reduces to the form of the generalized Hoek–Brown criterion at both triaxial and biaxial stress states. It has to be noted, however, that the proposed criterion is empirical in nature because it is derived from the 2D Hoek–Brown criterion, which is empirical. Because of the non-convexity of the yield surface for a biaxial stress state, the proposed criterion may have problems with some stress paths.
 4. The proposed generalized 3D Hoek–Brown strength criterion has been validated by applying it to polyaxial or true triaxial compression test data of intact rocks and jointed rock masses collected from the published literature.

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